Supporting Information

Beyond Candidate Generation: A Validation-First Protocol for AI-Assisted Transition-State Discovery

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S1. Protocol checklists

S1.1 P0: chemical preflight

- Reactive pairs: which bonds form and which break, with their actual distances in the starting geometry.
- Charge, multiplicity, and every accessible d^n spin state, including broken-symmetry solutions.
- Connectivity: build a bond graph with a distance criterion; check the intended bonds and any unplanned bond.
- Conformers: pre-fold flexible ligands/chains; retain several low-energy conformers.
- Reference state: reactant or encounter complex and product, at the same level as the TS.
- Candidate filter: hard overlaps (< 1.00 Å), clash ratios ($< 0.75\,\Sigma r_{\mathrm{cov}}$), heavy-atom RMSD, connectivity changes.

S1.2 P0.5: learned-potential pre-screen (optional)

The pre-screen sits between P0 and P1 and is the only stage that runs on a machine-learned potential (MLIP) instead of on DFT. It is optional, and no gate that decides whether a candidate is a transition state depends on it.

- Declare the reaction coordinate before screening: the forming and breaking pairs, the target bond lengths, the charge, and the multiplicity. The screen changes the geometry; it does not change the electronic-state assignment.
- Use the potential only inside its applicability domain. For broken-symmetry and open-shell transition-metal systems the verdict describes the learned surface, not the electronic state, and is recorded as a PES mismatch rather than as a basin.
- Displace the candidate along the declared coordinate in both directions and relax on the learned surface (the reference implementation defaults to 0.35 Å per side).
- Read the pair of endpoints. Same minimum on both sides: basin-like, discard before any DFT is spent. The two minima differ by exactly the declared forming and breaking bonds: barrier-like, pass the relaxed geometry to P1. Any other difference: the reactive pairs are wrong, and S1.1 is repeated before the screen is run again.
- PES mismatch: if the candidate lies more than 60 kcal mol^{-1} above both endpoints, the learned surface does not support this reaction. Do not read the verdict as a basin; skip the screen and go to P1 (rule 11 of Table 1 of the main text).

- Geometry repair: remove hard overlaps (0.83–0.93 Å in the cases reported here) and pull the declared bonds to their target lengths while the coordination sphere is held. Check that the coordination bonds are unchanged to within 0.05 Å before the geometry is used.
- Do not budget the repair as saved optimization cycles: ORCA projects the constraints onto the input geometry before the first cycle (measured, Ni–C* 3.000 → 2.150 Å), and the cycles that follow relax the ligand conformers.
- Record the model identity and version, the device, the declared pairs, and the verdict with the candidate, so that the pre-screen can be repeated or audited.

S1.3 P1: barrier-region scan

- Use typically 4–8 points along the reaction coordinate (more when the channel is shallow, has a double bump, or the first scan is inconclusive; the case study in the main text used 16), not a blind OptTS.
- Each point is an independent constrained optimization, starting from the previous converged geometry.
- Only converged points enter the profile; the highest converged point is the TS guess.
- Check the profile for monotonicity, shallow double bumps, and flat regions.

S1.4 P2: constrained reference and Hessian

- Run a constrained `Opt + Freq` at the chosen distance.
- The lowest modes must involve the declared reactive pair; otherwise do not run OptTS.
- Keep the Hessian and the reference geometry; both are reused downstream.

S1.5 P3: OptTS

- Read the P2 Hessian (`InHess Read` in ORCA; `readfc` in Gaussian).
- If displacing, scale by the target bond-length change (∼0.05 Å), not the maximum atomic displacement.
- Apply the stop rules collected in Table 1 of the main text.

S1.6 P4: validation

- Exactly one significant imaginary frequency.
- The imaginary mode moves the declared reactive pair; localization is not below 15%.
- Key bond lengths match the intended step.
- IRC or downhill relaxation reaches the intended reactant and product.
- $\langle S^2 \rangle$ and Mulliken spin are consistent with the geometry.
- Optimization/frequency/IRC at the same level; ΔE for minimum-vs-saddle; one solvation convention.

S2. Input templates

All templates use the level of theory of Section S3 unless stated. `%maxcore 1100` is a per-core limit (1100 MB), so the twelve-core examples need about 13 GB of scratch space; ORCA stops with an error when a per-core limit cannot be satisfied.

S2.1 ORCA: per-point constrained optimization

```
! Opt TPSS D3BJ def2-SVP RIJCOSX
%pal nprocs 12 end
%maxcore 1100
%cpcm
  smd true
  SMDsolvent "Toluene"
end
%geom
  MaxIter 220
  Constraints
    { B 20 56 1.750 C }
  end
end
* xyzfile 0 2 start.xyz
```

S2.2 ORCA: constrained reference + frequency

```
! Opt Freq TPSS D3BJ def2-SVP RIJCOSX
%pal nprocs 12 end
%maxcore 1100
%cpcm
  smd true
  SMDsolvent "Toluene"
end
%geom
  MaxIter 220
  Constraints
    { B 20 56 1.750 C }
  end
end
* xyzfile 0 2 start.xyz
```

S2.3 ORCA: OptTS reading the reference Hessian

```
! OptTS Freq TPSS D3BJ def2-SVP RIJCOSX
%pal nprocs 12 end
%maxcore 1100
%cpcm
  smd true
  SMDsolvent "Toluene"
end
%geom
  MaxIter 160
  InHess Read
  InHessName "L1-constOptFreq.hess"
```

```
end
* xyzfile 0 2 L1-constOptFreq.xyz
```

S2.4 ORCA: IRC

```
! IRC TPSS D3BJ def2-SVP RIJCOSX
%pal nprocs 12 end
%maxcore 1100
%irc
  MaxIter 20
  InitHess Read
  Hess_Filename "L2-TSopt.hess"
  Direction both
  PrintLevel 2
end
* xyzfile 0 2 L2-TSopt.xyz
```

S2.5 Gaussian: constrained reference + frequency

```
%mem=48GB
%nprocshared=24
%chk=01_const_optfreq.chk
#p UB3LYP/6-31G(d) opt=(calcfc,modredundant,maxcycles=220) freq
   em=gd3bj scrf=(smd,solvent=toluene) int=ultrafine

TS step 1: constrained Opt+Freq

0 2
<coordinates>

B 1 2 1.500 F
B 3 4 2.100 F
```

The level of theory in the Gaussian templates is an example; substitute the level used in the production runs (the case studies used TPSS-D3BJ/def2-SVP with SMD(toluene), see Section S3).

S2.6 Gaussian: OptTS reading the reference checkpoint

```
%mem=48GB
%nprocshared=24
%oldchk=01_const_optfreq.chk
%chk=02_ts_optfreq.chk
#p UB3LYP/6-31G(d) opt=(ts,readfc,noeigen,maxcycles=220) freq
   em=gd3bj scrf=(smd,solvent=toluene) int=ultrafine
   geom=check guess=read

TS step 2: OptTS+Freq

0 2
```

S2.7 Slurm template

```
#!/bin/bash
#SBATCH -J ts
#SBATCH -N 1
#SBATCH -n 24
#SBATCH -p <partition>
#SBATCH -o slurm-%j.out
#SBATCH --time=48:00:00

export g16root='<g16root>'
source "${g16root}/g16/bsd/g16.profile"
export GAUSS_SCRDIR=/tmp

date
bash run_chain.sh
echo "EXIT=$?"
date
```

S2.8 Learned-potential pre-screen and geometry repair

Both tools belong to the same package, run without a quantum-chemistry engine, and are called on an `xyz` candidate. The reactive pairs are given explicitly and the model is named on the command line, so that a reported verdict can be reproduced:

```
# basin-like / barrier-like verdict (pairs are 1-based)
python scripts/ts_mlip_screen.py \
    --cand candidate.xyz \
    --forming 21,72 --breaking 1,21 \
    --charge 0 --spin 2 \
    --model orb_v3_conservative_omol --device cuda

# geometry repair: i,j,d are 0-based, d in A
python scripts/mlip_preopt.py \
    --in raw.xyz --out repaired.xyz \
    --constraint 20,56,1.75 \
    --freeze-shell 1
```

`ts_mlip_screen.py` reports one of `BARRIER-LIKE`, `BASIN`, `WRONG-CHANNEL`, or `PES-MISMATCH` per candidate; with `--bs` (broken-symmetry singlet) it restricts itself to the geometry and issues no connectivity verdict. Indices are 1-based there and 0-based in `mlip_preopt.py`, which follows the convention of the ORCA `%geom Constraints` block. A run returns either a repaired geometry or a pair of relaxation endpoints; neither is a transition-state judgement, and the endpoints are read as described in S1.2.

S3. Case-study details

The systems in the main-text case studies are 84–98 atom Ni/Fe complexes with organic ligands, treated with TPSS-D3BJ/def2-SVP (RIJCOSX, SMD) for geometry and frequency calculations, and with higher-level single points where noted. The numbers quoted below are the observations discussed in the main text; the case numbers are those of Table 3 there. RIJCOSX is retained because it appears in the production inputs; for a non-hybrid functional the COSX approximation acts on no Hartree–Fock exchange.

S4. AI candidate generation: practical notes

S4.1 UniTS-Gen

The UniTS code package (which implements the UniTS-Gen model) takes `reactive_atom_idx` as 0-based, whereas many post-processing scripts use 1-based atom indices. Convert explicitly. The intended use is multi-sampling: generate ten candidates, prefilter, and send only a few to DFT. Single-shot correctness is moderate; the package's published benchmark reports that multi-sampling recovers all five previously failed cases. In an internal test on an 84-atom Ni broken-symmetry system, four candidates passed the geometric filter but had 3.6–4.4 Å heavy-atom RMSD to the reference; the model was out of distribution for that system class.

S4.2 ORCA GOAT-REACT

GOAT-REACT is a built-in, non-ML search. A minimal input is:

```
! GOAT-REACT B3LYP D3BJ def2-SVP TightSCF
%pal nprocs 12 end
%maxcore 1100
%geom
  Constraints
    { C 0 C }
    { C 4 C }
    { C 5 C }
  end
end
* xyz 0 1
<coordinates>
*
```

GOAT-REACT explores reaction pathways automatically; the reacting atoms can be held with Cartesian constraints in `%geom`, and ORCA counts atoms from zero. For Cartesian constraints the initial position is kept and no target value is given. No `%goat` block is used, so the defaults apply (its `MAXITER` key bounds the geometry iterations per worker, not the number of search steps). GOAT is a search tool; its output still goes through P2–P4. It becomes expensive for large flexible systems and is best used for channel discovery or medium-sized systems.

S5. Stop rules

The stop rules applied at each stage are collected in Table 1 of the main text; they are reproduced here only as a pointer so that the two documents cannot drift apart.